

Modulus equations and inequalities

Basic skills

Q1. a) $2x - 4 = 6 \Rightarrow 2x = 10 \Rightarrow x = 5$

b) $4 - 2x = 6 \Rightarrow -2x = 2 \Rightarrow x = -1$

c) solving $|2x - 4| = 6$ is equivalent to solving

$$2x - 4 = 6$$

$$4 - 2x = 6$$

if $2x - 4 > 0$ i.e. $x > 2$

if $2x - 4 < 0$ i.e. $x < 2$

solving $x = 5$

solving $x = -1$

$5 > 2$ so yes

$-1 < 2$ so yes

important
to consider
limits

$x = -1, 5$

Q2. $|6x + 2| = 10$

solve

$$6x + 2 = 10$$

and

$$-6x - 2 = 10$$

if $x > -\frac{1}{3}$

if $x < -\frac{1}{3}$

solving

$$x = \frac{8}{6} = \frac{4}{3} > -\frac{1}{3}$$

so yes $x = \frac{4}{3}$

solving

$$x = \frac{12}{-6} = -2 < -\frac{1}{3}$$

so yes $x = -2$

Q3. a) $5x - 4 > 6 \Rightarrow 5x > 10 \Rightarrow x > 2$

b) $4 - 5x > 6 \Rightarrow -5x > 2 \Rightarrow x < -\frac{2}{5}$

$\div -5$ so change sign
 $> \rightarrow <$

Q4. $|5 - 2x| = x$

solve $5 - 2x = x$

if $x < \frac{5}{2}$

$$3x = 5$$

$$x = \frac{5}{3} < \frac{5}{2}$$

so $x = \frac{5}{3}$

$2x - 5 = x$

if $x > \frac{5}{2}$

$$x = 5 > \frac{5}{2}$$

so $x = 5$

two solutions

$$x = \frac{5}{3}, 5$$

Competency skills

Q5. a) $|3x+2| = 5$

$$3x+2=5$$

$$\text{if } x > -\frac{2}{3}$$

$$x = 1 > -\frac{2}{3}$$

$$-3x-2=5$$

$$\text{if } x < -\frac{2}{3}$$

$$x = -\frac{7}{3} < -\frac{2}{3}$$

$$x = -\frac{7}{3}, 1$$

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b) As in prior part $x > -\frac{2}{3}$, $x < -\frac{2}{3}$

c) $3x+2 < 5$

$$x < 1$$

so combine with

$$x > -\frac{2}{3}$$

$$-\frac{2}{3} < x < 1$$

As required

$$-3x-2 < 5$$

$$x > -\frac{7}{3}$$

so combine with

$$x < -\frac{2}{3}$$

$$-\frac{7}{3} < x < -\frac{2}{3}$$

d) for $|3x+2| > 5$

$$3x + 2 > 5$$

$$x > 1$$

so with $x > -\frac{2}{3}$

$$\text{get } x > 1$$

$$-3x - 2 > 5$$

$$x < -\frac{7}{3}$$

with $x < -\frac{2}{3}$

$$\text{get } x < -\frac{7}{3}$$

$$\text{so } x > 1 \text{ or } x < -\frac{7}{3}$$

$$\text{Q6. } |6x - 4| > 2x$$

$$6x - 4 > 2x$$

with $6x - 4 > 0$

$$\text{so } x > \frac{2}{3}$$

solving

$$4x - 4 > 0$$

$$x > 1$$

with $x > \frac{2}{3}$

$$\text{so } x > 1$$

$$4 - 6x > 2x$$

with $4 - 6x < 0$

$$\text{so } x < \frac{2}{3}$$

solving

$$4 > 8x$$

$$x < \frac{1}{2}$$

with $x < \frac{2}{3}$

$$\text{so } x < \frac{1}{2}$$

$$\text{so } x < \frac{1}{2} \text{ or } x > 1$$

Q7. $|\tan x| = 1$ $-\pi < x \leq \pi$

$$\tan x = 1$$

if $\tan x > 0$

$$0 < x < \frac{\pi}{2}$$

$$-\pi < x < -\frac{\pi}{2}$$

$$x = \tan^{-1} 1 = \frac{\pi}{4}$$

$$\frac{\pi}{4} \text{ or } -\frac{3\pi}{4}$$

In domain so

$$x = -\frac{3\pi}{4}, \frac{\pi}{4}$$

$$-\tan x = 1$$

if $\tan x < 0$

$$\frac{\pi}{2} < x < \pi$$

$$-\frac{\pi}{2} < x < 0$$

$$x = \tan^{-1}(1) = -\frac{\pi}{4}$$

$$-\frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

In domain so

$$x = -\frac{\pi}{4}, \frac{3\pi}{4}$$

$$x = -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}$$

Q8. $|2x-1| = 2|x|-1$ $\&$ At $(2x-1)^2 = (2|x|-1)^2$

so need 4 cases (2 for each modulus)

- $2x-1 = 2x-1$ $\&$ true for $x \in \mathbb{R}$

if $2x-1 > 0$, $x > 0$ i.e. $x > \frac{1}{2}$

solving is true for any x i.e. $x \in \mathbb{R}$
 $x \in \mathbb{R}, x > \frac{1}{2}$ gives $x > \frac{1}{2}$

- $2x - 1 = -2x - 1$

if $2x - 1 > 0, x < 0$ so this case is impossible $\rightarrow x > \frac{1}{2} \text{ \& } x < 0$

- $-2x + 1 = 2x - 1$

if $2x - 1 < 0, x > 0$ i.e. $0 < x < \frac{1}{2}$

solving $4x = 2$ $x = \frac{1}{2}$ not in domain so no solution

- $-2x + 1 = -2x - 1$

if $2x - 1 < 0, x < 0$ i.e. $x < 0$

solving is never true so no solution.

conclusion - $x > \frac{1}{2}$ is only region

Q9. $|e^{4x-2}| = |-2|$

$e^{4x-2} = 2$ ↙ always +ve ↘ make +ve

$$4x - 2 = \ln 2$$

$$x = \frac{2 + \ln 2}{4}$$

Advanced skills

Q10. a) $|\cos x| = \frac{1}{2} \quad -\pi < x \leq \pi$

$$\cos x = \frac{1}{2}$$

$$-\cos x = \frac{1}{2}$$

if $\cos x > 0$

i.e. $-\frac{\pi}{2} < x < \frac{\pi}{2}$

if $\cos x < 0$

i.e. $\frac{\pi}{2} < x \leq \pi$ or

$-\pi < x < -\frac{\pi}{2}$

$$x = \cos^{-1} \frac{1}{2} = \frac{\pi}{3}$$

$$x = \cos^{-1} \left(-\frac{1}{2} \right) = \frac{2\pi}{3}$$

so $\frac{\pi}{3}, -\frac{\pi}{3}$

$\frac{2\pi}{3}, -\frac{2\pi}{3}$

so $x = -\frac{2\pi}{3}, -\frac{\pi}{3}, \frac{\pi}{3}, \frac{2\pi}{3}$

b) $|\cos x| \leq \frac{1}{2}$

$$\cos x \leq \frac{1}{2}$$

in $-\frac{\pi}{2} < x < \frac{\pi}{2}$

solving

$$x \geq \frac{\pi}{3} \text{ or } x \leq -\frac{\pi}{3}$$

so

(1a) $\frac{\pi}{3} \leq x < \frac{\pi}{2} \text{ or }$

(1b) $-\frac{\pi}{2} < x \leq -\frac{\pi}{3}$

$$-\cos x \leq \frac{1}{2}$$

in $\frac{\pi}{2} < x \leq \pi \text{ or }$

$$-\pi < x < -\frac{\pi}{2}$$

solving

$$\cos x \geq -\frac{1}{2}$$

$$-\frac{2\pi}{3} \leq x \leq \frac{2\pi}{3}$$

so

(2a) $\frac{\pi}{2} < x \leq \frac{2\pi}{3} \text{ or }$

(2b) $-\frac{2\pi}{3} \leq x < -\frac{\pi}{2}$

Then combining as either case

$$\frac{\pi}{3} \leq x \leq \frac{2\pi}{3} \text{ or } -\frac{2\pi}{3} \leq x \leq -\frac{\pi}{3}$$

(1a) + (2a)

(1b) + (2b)

A better approach is graphically



Alternatively
 $(2-4x)^2 = (4|x+2|-1)^2$

Q11. $|2-4x| = 4|x+2|-1$

- $2-4x = 4(x+2)-1$

$$2-4x > 0, x+2 > 0 \quad \text{i.e.} \quad x < \frac{1}{2}, x > -2$$

$$\text{so } -2 < x < \frac{1}{2}$$

solving $8x = -5 \quad x = -\frac{5}{8}$ in domain
 so yes

- $2-4x = -4(x+2)-1$

$$2-4x > 0, x+2 < 0 \quad \text{i.e.} \quad x < \frac{1}{2}, x < -2$$

$$\text{so } x < -2$$

solving $2-4x = -9-4x$
 no solution

- $4x - 2 = 4(x+2) - 1$

$$2 - 4x < 0, \quad x + 2 > 0 \quad \text{i.e.} \quad x > \frac{1}{2}, \quad x > -2$$

$$\text{so } x > \frac{1}{2}$$

$$\text{solving } 4x - 2 = 4x + 7$$

no solution

- $4x - 2 = -4(x+2) - 1$

$$2 - 4x < 0, \quad x + 2 < 0 \quad \text{i.e.} \quad x > \frac{1}{2}, \quad x < -2$$

so no domain

$$\text{Conclusion: } x = -\frac{5}{8}$$

Q12. $|3x - 2| + 2x > 5 - x$

$$|3x - 2| > 5 - 3x$$

- $3x - 2 > 5 - 3x$

$$\text{with } 3x - 2 > 0 \quad \text{so } x > \frac{2}{3}$$

$$\text{solving } 6x > 7 \quad x > \frac{7}{6} \text{ in domain}$$

so $x > \frac{7}{6}$

• $2 - 3x > 5 - 3x$

with $3x - 2 < 0$ so $x < \frac{2}{3}$

solving gives no solutions (as $2 > 5$)

Conclusion - $x > \frac{7}{6}$

Q13. $|x^2 + b| = c$

case 1:

$x^2 + b = c$

if $x^2 + b > 0$ i.e. $x^2 > -b$

$x^2 = c - b$

$x = \pm \sqrt{c - b}$

need $c - b \geq 0$

$c \geq b$

need $c - b > -b$ so $c > 0$

Two if $c > b$, $c > 0$

one if $c = b$, $c > 0$

Case 2: $-x^2 - b = c$ if $x^2 + b < 0$

so $b < -x^2$
 $x^2 = -b - c$

$x = \pm \sqrt{-(b+c)}$ need $b+c \leq 0$

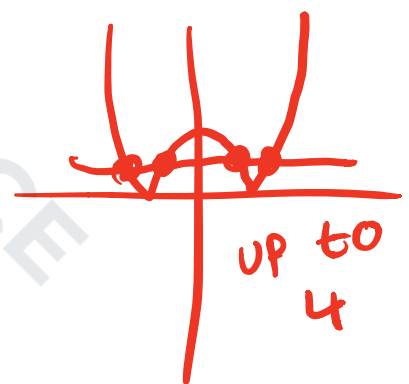
need $-x^2 > b$
 $b+c > b$ so $c > 0$

Two if $b+c < 0$, $c > 0$

One if $c = -b$, $c > 0$

So 4 solutions at most

e.g. $|x^2 - 2| = 1$



Q14. $||x|^2 - 4|x|| = 2$

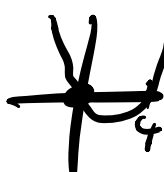
$\checkmark |x|^2 = x^2$ as $x^2 \geq 0$

so $|x^2 - 4|x|| = 2$

Alt $(x^2 - 4|x| - 1)^2 = 4$

split into cases:

• $x^2 - 4x = 2$

if $x \geq 0$ and $x^2 - 4x > 0$ 

$x < 0$ or $x > 4$

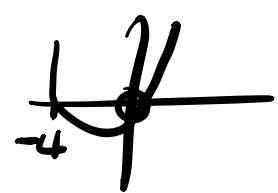
so that is if $x > 4$

then solving $x^2 - 4x - 2 = 0$
gives

$$x = \frac{4 \pm \sqrt{16 + 8}}{2} = 2 \pm \sqrt{6}$$

$2 + \sqrt{6} > 4$ so $x = 2 + \sqrt{6}$ is
a solution

• $x^2 + 4x = 2$

if $x \leq 0$, $x^2 + 4x > 0$ 

$x < -4$ or $x > 0$

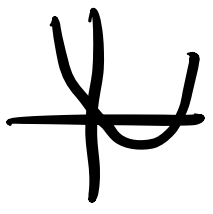
so $x < -4$

solving $x^2 + 4x - 2 = 0$

$$x = \frac{-4 \pm \sqrt{16 + 8}}{2} = -2 \pm \sqrt{6}$$

$-2 - \sqrt{6} < -4$ so $x = -2 - \sqrt{6}$ is a solution

• $-(x^2 - 4x) = 2$

if $x \geq 0$, $x^2 - 4x < 0$ 
 $0 < x < 4$

so $0 < x < 4$

solving

$$x^2 - 4x + 2 = 0$$

$$x = \frac{4 \pm \sqrt{16 - 8}}{2} = 2 \pm \sqrt{2}$$

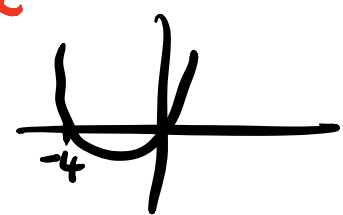
both in domain

$$\text{so } x = 2 \pm \sqrt{2}$$

• $-(x^2 + 4x) = 2$

if $x \leq 0$

$$x^2 + 4x < 0$$

$-4 < x < 0$ 

so $-4 < x < 0$

solving $x^2 + 4x + 2 = 0$

$$x = \frac{-4 \pm \sqrt{16-8}}{2}$$

$$= -2 \pm \sqrt{2}$$

both in domain

so $x = -2 \pm \sqrt{2}$

conclusion:

$$x = 2 + \sqrt{6}, -2 - \sqrt{6}, 2 \pm \sqrt{2}, -2 \pm \sqrt{2}$$